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Course code :- CSA0904

Reg no :- 192372075

Course Name :- Programming For
Java

1. Student management system:

Class student

PRIVATE name, roll NO, marks

METHOD accept Details()

INPUT Name roll NO, marks

END METHOD

METHOD calculate Average()

Return marks / subjects

END METHOD

METHOD display Result()

IF calculate Average() ≥ 40

Print "Pass"

ELSE

Print "Fail"

END METHOD

END class

2. Library Book Management

Class Book

PRIVATE title, author, ISBN, is-Available

METHOD issue_Book()

is Available = FALSE;

END METHOD

METHOD, return Book()

is - Available = TRUE

END METHOD.

```
METHOD displayDetailsc )
```

```
PRINT title, author, ISBN, is Available
```

```
END method.
```

```
END class
```

3. Employee salary calculation:-

```
Class Employee
```

```
Private EMPID, Name, basicpay
```

```
method calculateGrossC )
```

```
Return basicpay + HRA + DA
```

```
END method
```

```
method calculateNetC )
```

```
Return calculateGrossC() - PF - Tax
```

```
END method
```

```
method displayPaySlipC )
```

```
Print calculateNetC )
```

```
END method
```

```
END class
```

4. Shape Area calculator:

```
class Shape
```

```
method calculateAreaC )
```

```
END class
```

```
class Circle EXTENDS Shape
```

```
method calculateAreaC )
```

```
Return 3.14 * r * r
```

END class

class Rectangle Extends Shape

Method calculate Area()

Return $l * b$

END class

5. Vehicle Registration System.

Class vehicle

regno; owner, fuel type

Method display details()

END class

Class Two wheeler Extends vehicle

Engine capacity.

END class.

6. Bank Account Hierarchy:

Class BankAccount

acc no, holder, balance.

END class

class Savings Account Extends Bank account

Method withdraw (amount)

IF amount $\leq$ balance

balance = amount

END class

class current Account Extends Bankaccount

METHOD withdraw (amount).

IF amount $\leq$ balance + overdrafts

balance = amount

END class

7. Hospital Record System:

class Patient

Private diagnosis, Prescription

Method generate Bill()

END class

class Inpatient Extends Patient

Method generate Bill()

Return days * daily charge

END class

8. E-Commerce Product Catalog:

Class Product

Method apply Discount()

END class

Class Electronics Extends Product

Method apply Discount()

Return price * 0.90

END class

Class Clothing Extends Product

Method apply Discount()

Return price * 0.80

END class

Class Grocery Extends Product

Method apply Discount()

Return Price * 0.9

END class.

9. University staff hierarchy

class staff

Private salary

Protected Name

Method Calculate Allowance()

END class

class teaching staff extends staff

END class

class professor extends teaching staff

END class.

10. Smart Home Device Controller.

class Smart Device

Method execute command(cmd)

END class

class smart light extends smart Device

Method execute command(cmd)

Print "Light Action"

END class

class smartlock extends smart Device

private pin

Method execute command(cmd)

END class.